

Acid–Base Titration Curves

Answer key with worked solutions

A. Understanding

1. Equivalence point vs end point.

The **equivalence point** is where the moles of base added exactly match the moles of acid (from the balanced equation). The **end point** is where the indicator changes colour. A good indicator makes them almost the same.

2. Equivalence-point pH.

(a) **Equal to 7**: strong acid + strong base. (b) **Above 7**: weak acid + strong base (formate is weakly basic). (c) **Below 7**: strong acid + weak base (NH_4^+ is weakly acidic).

B. Strong acid with strong base

3. Equivalence volume.

Moles HCl = $0.150 \text{ mol/L} \times 0.02000 \text{ L} = 0.00300 \text{ mol}$. Volume NaOH = $0.00300 \text{ mol} \div 0.100 \text{ mol/L} = 0.0300 \text{ L} = \mathbf{30.0 \text{ mL}}$

4. Initial pH.

$\text{pH} = -\log(0.150) = \mathbf{0.82}$

5. After 10.00 mL NaOH.

NaOH added = $0.100 \text{ mol/L} \times 0.01000 \text{ L} = 0.00100 \text{ mol}$; HCl left = $0.00300 \text{ mol} - 0.00100 \text{ mol} = 0.00200 \text{ mol}$ in 30.00 mL. $[\text{H}_3\text{O}^+] = 0.00200 \text{ mol} \div 0.03000 \text{ L} = 0.0667 \text{ M}$; $\text{pH} = \mathbf{1.18}$

6. After 40.00 mL NaOH.

NaOH added = $0.100 \text{ mol/L} \times 0.04000 \text{ L} = 0.00400 \text{ mol}$; excess $\text{OH}^- = 0.00400 \text{ mol} - 0.00300 \text{ mol} = 0.00100 \text{ mol}$ in 60.00 mL. $[\text{OH}^-] = 0.00100 \text{ mol} \div 0.06000 \text{ L} = 0.0167 \text{ M}$; $\text{pOH} = 1.78$; $\text{pH} = 14.00 - 1.78 = \mathbf{12.22}$

C. Weak acids and indicators

7. pKa and Ka from the half-equivalence point.

At half-equivalence, pH = pKa, so **pKa = 4.20** and $K_a = 10^{-4.20} = \mathbf{6.3 \times 10^{-5}}$

8. Indicator choice.

Phenolphthalein (pH 8.2–10.0). The equivalence point of a weak acid–strong base titration is above 7, inside the steep jump where phenolphthalein changes colour. Methyl orange would change far too early.